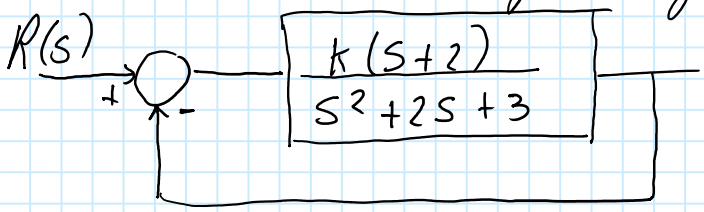
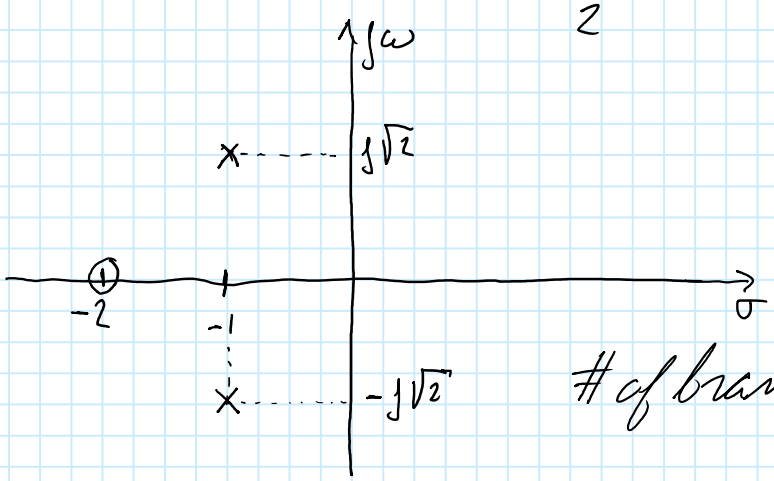
1) Sketch the root locus of the following system



a) Locate poles and zeros / # of branches

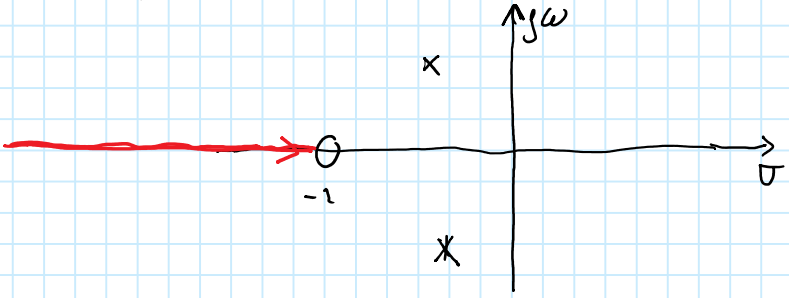
$z_1 = -2$; $s^2 + 2s + 3 = 0$

$$\frac{-2 \pm \sqrt{4-4(1)(3)}}{2} \rightarrow -1+j\sqrt{2}, \; -1-j\sqrt{2}$$



of branches = n = 2

b) Root locus on real axis



c) Asymptotes:

of asymptotes = n - m = 2 - 1 = 1

$$\angle \text{ asymptotes} = \pm \frac{180(2k+1)}{2-1} = \pm 180(2k+1)$$

$k = 0 \Rightarrow \angle = \pm 180$ ($+180°$ and $-180°$ are the same asymp.)

$\angle$ asymptote $= 180°$ (it's the one on real axis)

180° means a horizontal axis, the one on the real axis.

d) centroid (intersection of asymptotes)

$$\sigma_c = \frac{(-1+j\sqrt{2}) + (-1-j\sqrt{2}) - (-2)}{2-1}$$

$$= \frac{-2+2}{1} = 0$$

There's only one asymptote, so there's not really an intersection.

e) Breakaway, break-in point

$$1+GH = 1 + \frac{K(s+2)}{s^2+2s+3} = 0$$

$$\frac{K(s+2)}{s^2+2s+3} = -1 \; ; \; K = -\frac{(s^2+2s+3)}{s+2}$$

$$\frac{dK}{ds} = -\left[\frac{(2s+2)(s+2) - (1)(s^2+2s+3)}{(s+2)^2}\right] = 0$$

$$2s^2+4s+2s+4 - s^2-2s-3 = 0$$

$$2s^2+6s+4-s^2-2s-3 = 0$$

$$s^2+4s+1 = 0$$

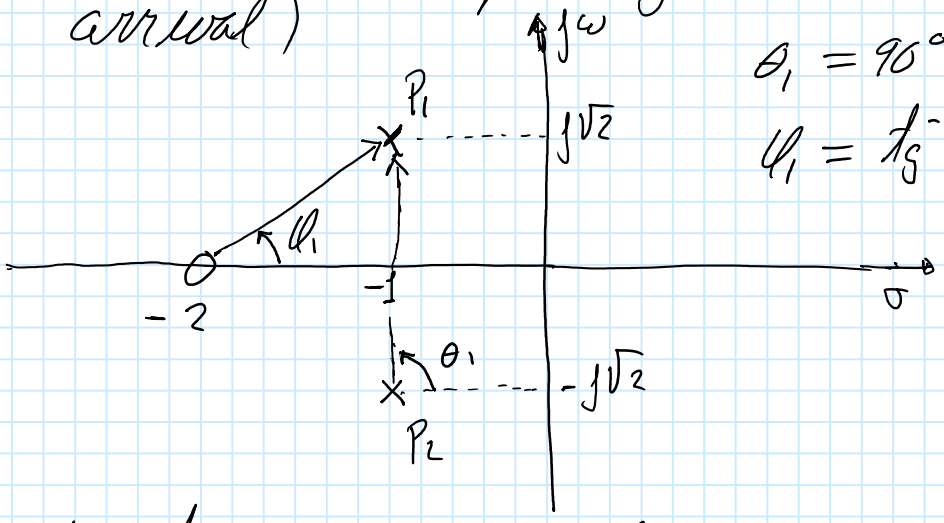
$$\frac{-4 \pm \sqrt{4^2-4(1)(1)}}{2(1)} = \frac{-4 \pm \sqrt{12}}{2} = -1 \pm \sqrt{3}$$

$-1+\sqrt{3}$ does not belong to the R.L

$-1-\sqrt{3} \simeq -2.732$ belongs to the R.L ✓

it's a break-in point since it's between 2 zeros (-2 and $-\infty$)

f) angle of departure from complex poles (there's not complex zeros ⇒ no angle of arrival)



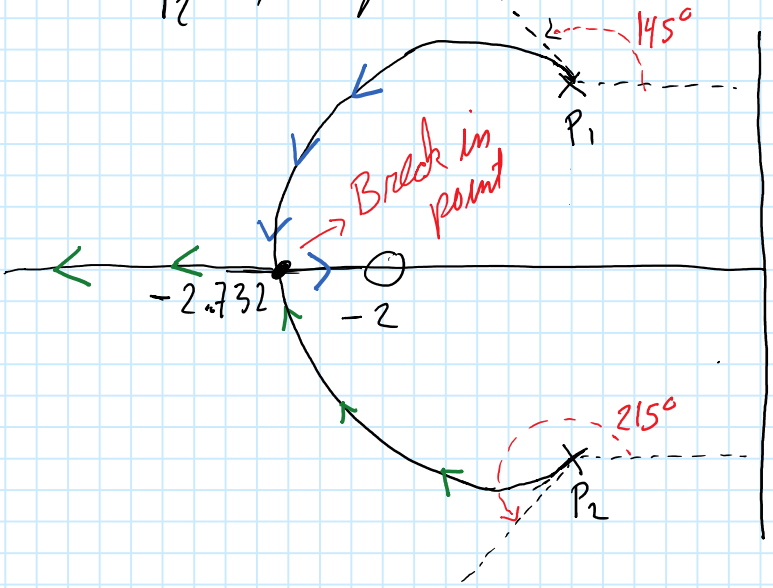
$\theta_1 = 90°$

$\phi_1 = \tan^{-1}\frac{\sqrt{2}}{2-1} = 54.7356°$

P_1 ∠ departure $= 180 - \theta_1 + \phi_1 = 180 - 90 + 54.74° = 144.74° \approx 145°$

because of symmetry

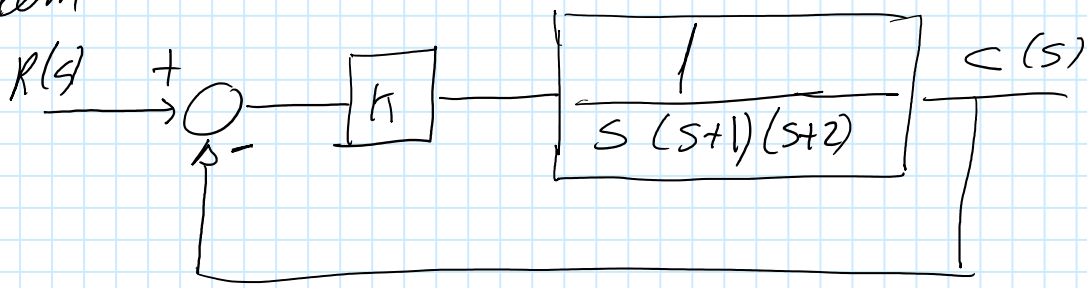
P_2 ∠ departure $= 360 - 145 = 215°$



g) Intersection with jω axis → None as no R.L. goes in that direction

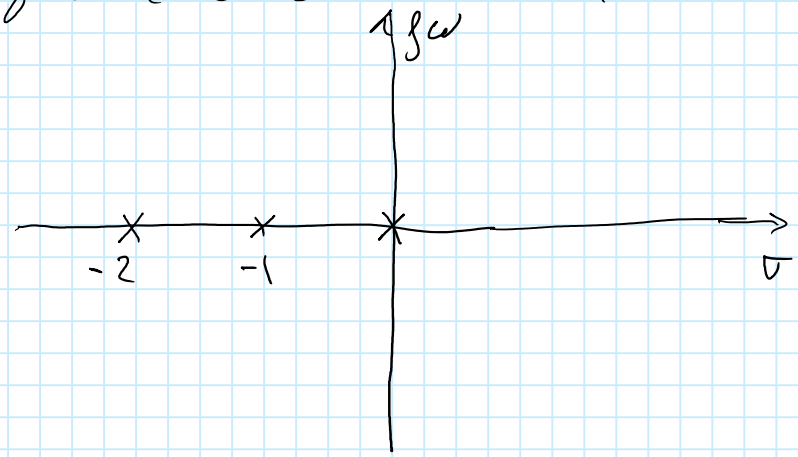
∴ the system is always stable for all $K > 0$

2) Sketch the root locus of the following system

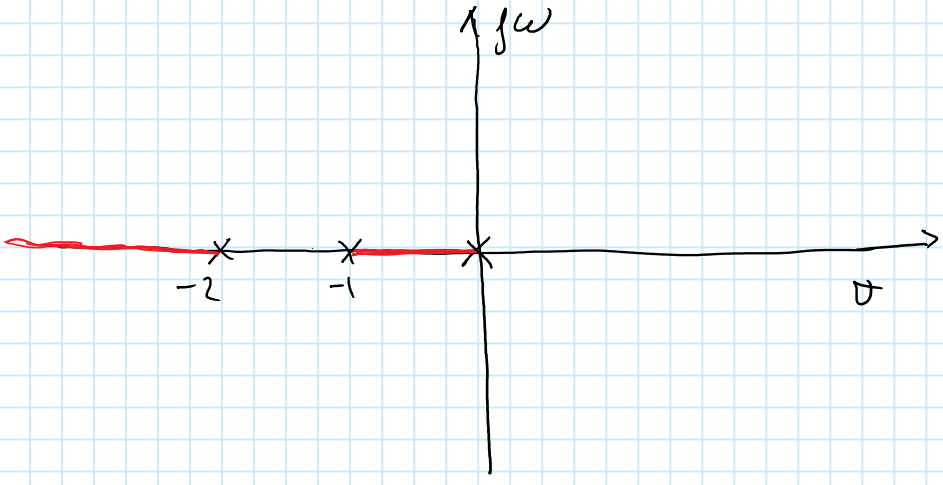


a) Pole-Zero diagram, # of branches

of branches = 3 = n



b) R.L. on real axis



c) Asymptotes

$$\#\ \text{asym.} = n - m = 3 - 0 = 3$$

$$\angle\ \text{asym.} = \frac{\pm 180(2k+1)}{n-m} = \pm 60(2k+1) \qquad k = 0, 1, 2, \ldots$$

$k = 0 \rightarrow +60°$, $-60°$

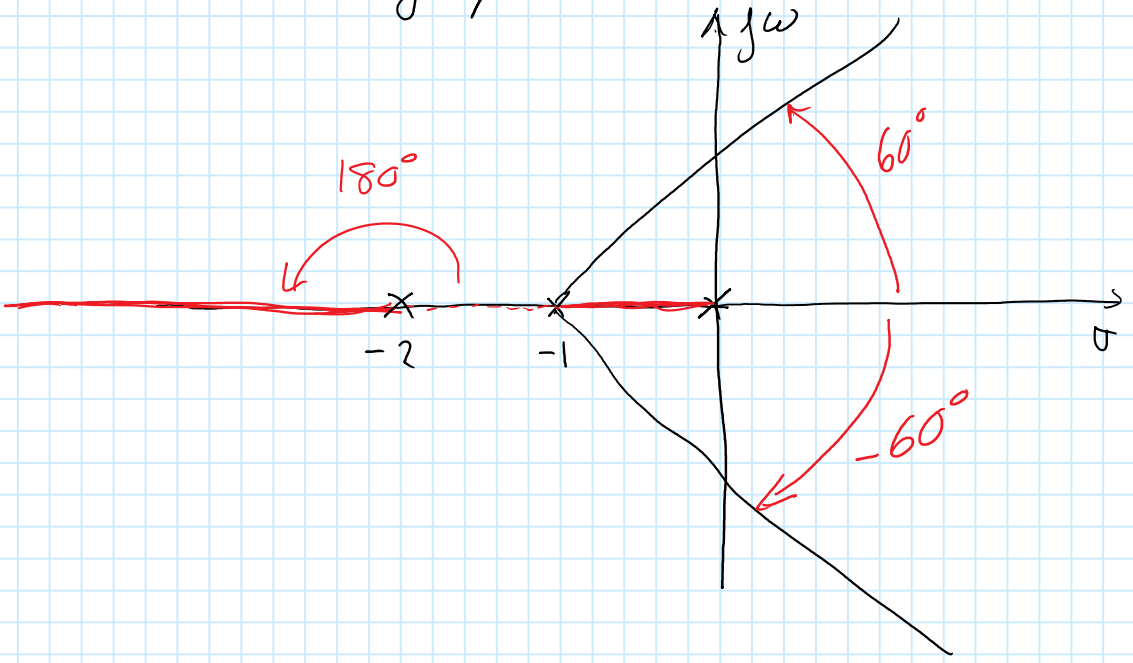
$k = 1 \rightarrow \pm 180$ (+180 and -180 are equivalent)

$\angle\ \text{asym.} = 60, -60, +180$

d) Centroid

$$\sigma_c = \frac{\sum \text{poles} - \sum \text{zeros}}{n-m} = \frac{(0-1-2)+(0)}{3} = -1$$

all asymptotes intersect @ $\sigma_c = -1$



d) Breakaway/Break-in point

Between $p_1 = 0$ and $p_2 = -1$ there must be a breakaway point.

$$1 + GH = 1 + \frac{K}{s(s+1)(s+2)} = 0$$

$$\frac{K}{s(s+1)(s+2)} = -1 \Rightarrow K = -s(s+1)(s+2)$$

$$\frac{dK}{ds} = -\left[(s+1)(s+2) + s(s+2) + s(s+1)\right]$$

$$= -\left[s^2+3s+2 + s^2+2s + s^2+s\right] = -\left[3s^2+6s+2\right] =$$

$$-(3s^2+6s+2) = 0 \Rightarrow 3s^2+6s+2 = 0$$

$$\frac{-6 \pm \sqrt{6^2-4(3)(2)}}{2(3)} = -1 \pm \frac{\sqrt{3}}{3} \begin{cases} \rightarrow -0.42265 \\ \rightarrow -1.57735 \end{cases}$$

$s_1 = -0.42265$ belongs to the R.L. since is between $s=0$ an $s=-1$ ✓

$s_2 = -1.57735$ does not belong to R.L.

e) angle of departure/arrival

no angle of departure/arrival since there are no complex zeros/poles.

f) intersection with the $j\omega$ axis

In that case there are 2 pure imaginary poles

$$s_{1,2} = \pm j\omega$$

let's take $s = j\omega$

$$1 + GH(j\omega) = 0$$

$$1 + \frac{K}{(j\omega)(j\omega+1)(j\omega+2)} = 0$$

$$j\omega(j\omega+1)(j\omega+2) + K = 0$$

$$(-\omega^2 + j\omega)(j\omega+2) + K = 0$$

$$(-j\omega^3 - 2\omega^2 - \omega^2 + j2\omega) + K = 0$$

$$(-3\omega^2 + K) + j(2\omega - \omega^3) = 0$$

$$\Rightarrow -3\omega^2 + K = 0 \quad (1)$$

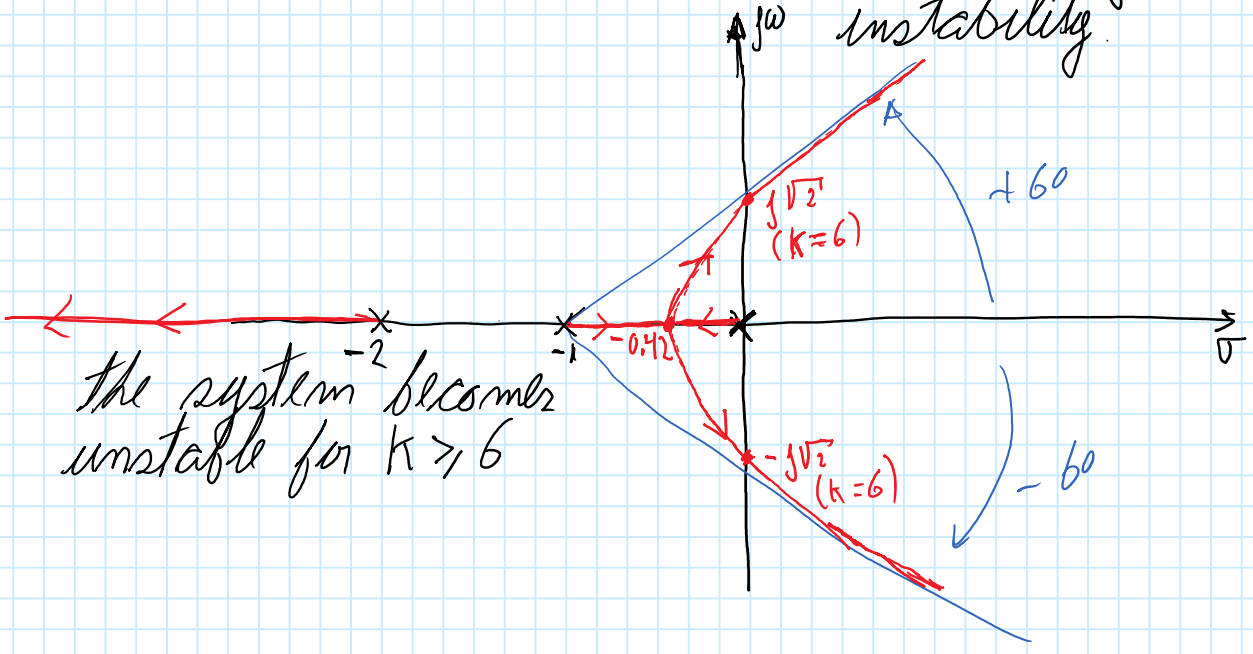
$$2\omega - \omega^3 = 0 \Rightarrow 2 - \omega^2 = 0 \Rightarrow \omega_{1,2} = \pm\sqrt{2}$$

$$\omega = \sqrt{2}$$

Use (1):

$$-3(\sqrt{2})^2 + K \Rightarrow K = 6$$

value of K when poles cross the $j\omega$ axis to instability.



the system becomes unstable for $K \geq 6$